Paired quantum dilogarithms: source dictionary and bounded kernel tests

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Abstract

Existing Gaussian-group Fourier identities offer a direct language for the bounded paired pentagon. We match their measure and sign conventions, distinguish three nearby constructions, and give a heat-regularized kernel test without unbounded chiral factors. Its matrix-element error reduces to a positive Gaussian phase-loss expectation. An empirical Sobol study explores the size of that functional, but does not certify it or evaluate the pentagon defect. The original bounded identity and the physical observer gluing problem remain open in this project.

1 Literature first: what can be imported

The scalar function under study is the Kim–Scarinci (KS) phase [1]

$$F_\hbar(x,y) = \frac{\Phi^{i\hbar}(x+i\hbar y)}{\overline{\Phi^{i\hbar}(x+i\hbar y)}}, \qquad \hbar > 0.$$

It has modulus one almost everywhere, with arbitrary values assigned at the discrete exceptional points. Coordinates are dimensionless. The claim being reproduced is KS Proposition 6.15, in the representation

$$Q = (X,Y), \quad P = (P_y, P_x), \quad P_j = -\pi i \partial_j, \quad A = F_\hbar(Q), \quad B = F_\hbar(P), \quad M = e^{iXY/\pi},$$

$$AB = BM^{-1}BMA. \tag{1}$$

The components of each vector strongly commute. The shift in the middle factor is the self-adjoint closure specified by $M^{-1}PM = P + Q$.

Garoufalidis–Kashaev (GK), Appendix B, equations (136)–(138) [2], provides the bounded-operator/Fourier-kernel equivalence on a Gaussian group. GK's later Remark 2.6 restates the distributional version [3]. These results give a framework, not an automatic proof that our particular function satisfies its required identity. Kashaev's earlier Lemma 1 and Theorem 1 have square-integrability hypotheses [4]; a unit-modulus function on $\mathbb{R}^2$ is not in $L^2(\mathbb{R}^2)$. A damping factor supplies integrability but generally destroys the exact pentagon, so it cannot be inserted into those theorems as though the hypothesis were unchanged.

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2 The precise Gaussian group and nearby examples

Use the additive group $G = \mathbb{R}^2$ with

$$\gamma(s) = e^{-is_x s_y/\pi}, \qquad \chi(s,t) = e^{-i(s_x t_y + s_y t_x)/\pi}, \qquad d\mu(s) = \frac{ds_x\, ds_y}{2\pi^2}.$$

The negative-sign Fourier transform S with kernel χ is unitary, $S^2 f(s) = f(-s)$, and its Gaussian integral is 1 in the oscillatory sense. Thus $SQS^* = -P$. The source convention is $f(q) = f(Q)$, $f(p) = Sf(Q)S^* = f(-P)$ and $f(p+q) = \gamma(Q)^* f(p)\gamma(Q) = f(Q-P)$. Conjugating the source pentagon by S^* gives exactly (1).

With $k = S^* F_\hbar$ as a tempered distribution, the matched version of GK's equation (137) is

$$k(a)k(b)\chi(a,b) = \int_G k(b-c)k(c)k(a-c)\gamma(c)\, d\mu(c). \tag{2}$$

Equation (2) is the identity to establish, not an ordinary absolutely convergent integral already known for this k.

The known real-plane example. GK Example 2.4 uses $\varphi_b(\dot{x} + \ddot{x}/2)/\varphi_b(\dot{x} - \ddot{x}/2)$, with the usual Faddeev convention. For real $b = \sqrt{\hbar}$, set

$$\dot{x} = \frac{x}{2\pi\sqrt{\hbar}}, \qquad \ddot{x} = -\frac{\sqrt{\hbar}\, y}{\pi}.$$

The absolute Jacobian is $1/(2\pi^2)$ and $e^{2\pi i \dot{x}\ddot{x}} = \gamma(x,y)$. Since the KS convention is $\Phi^\hbar(z) = \varphi_{\sqrt{\hbar}}(z/(2\pi\sqrt{\hbar}))^{-1}$, this gives the KS *negative*-cosmological-constant pairing

$$\frac{\Phi^\hbar(x+\hbar y)}{\Phi^\hbar(x-\hbar y)}.$$

Replacing $\hbar$ by $i\hbar$ is not a real change of Haar coordinates. The real-parameter ratio is smooth on the real plane, whereas our phase has isolated nonremovable windings. Thus a real invertible linear substitution, even with a smooth nonvanishing phase, cannot identify these two multipliers. This does not exclude a more general unitary equivalence.

The local-field construction. GK Definition 3.1 and Theorem 3.3 use $B = F^\times$, $G = \widehat{B} \times B$, and $\varphi(\alpha, z) = \alpha(1+z)$. For $F = \mathbb{C}$ this group is $\mathbb{R}^2 \times \mathbb{Z} \times \mathbb{T}$, not the additive complex plane. Lemma 3.2 gives a Fourier expression away from $z = 1$ and explicitly retains its distributional qualification there. The theorem does not identify this construction with $F_\hbar$.

3 A scalar cylinder dictionary is not an operator dictionary

For real $0 < q < 1$, write

$$C_q(\theta, m) = \frac{(-q^{1-m}e^{i\theta}; q^2)_\infty}{(-q^{1-m}e^{-i\theta}; q^2)_\infty}.$$

For integer m, this is the $\mathbb{T} \times \mathbb{Z}$ phase in GK Appendix A, equations (129)–(131) [2]. Away from exceptional arguments, the same product is a scalar function of real m. The KS product formula gives the exact scalar factorization

$$F_\hbar(x, y) = \frac{C_{\widetilde{q}}(-x/\hbar, \hbar y/\pi)}{C_q(\hbar y, x/(\pi\hbar))}, \qquad q = e^{-\pi\hbar}, \quad \widetilde{q} = e^{-\pi/\hbar}. \tag{3}$$

Both second arguments in (3) are generally noninteger.

The known inversion formula

$$C_q(\theta, m)C_q(-\theta, -m) = e^{im\theta}, \qquad m \in \mathbb{Z},$$

cannot hold for arbitrary real m as a function on the circle. The left side is 2π-periodic, whereas the proposed right side has monodromy $e^{2\pi i m}$. Passing to the real universal cover changes the group and its Fourier dual; it does not extend the original operator theorem without further work. Likewise, integer radial spectra in the quantum-$az + b$ theory cannot silently be replaced by the full complex spectrum of $e^{X+i\hbar Y}$.

The *combined* phase in (3) still has the correct inversion,

$$F_\hbar(x, y)F_\hbar(-x, -y) = e^{-ixy/\pi}. \tag{4}$$

It follows directly from KS's scalar inversion formula, including its real constant at parameter $i\hbar$. Thus the individual continuous-index failure does not prove a failure of the paired pentagon. Sixty-digit checks of nine generic $(\hbar, x, y)$ points verify (3) and (4); four integer cylinder controls and three noninteger monodromy controls are also saved.

What the search has established. There is an applicable *formulation* of the bounded identity. The real-plane, cylinder, and local-field examples do not yet supply its missing proof for our phase in this representation. These are identified hypothesis differences, not a claim that all relevant literature has been exhausted. We do not claim a new general quantum-dilogarithm theorem.

4 An absolutely defined regulated kernel

Let

$$H_\epsilon = e^{-\epsilon(P_x^2+P_y^2)}, \quad B_\epsilon = BH_\epsilon, \quad k_\epsilon = S^*(e^{-\epsilon|s|^2}F_\hbar(s)), \qquad \epsilon > 0.$$

The Fourier integral defining k_ϵ is absolutely convergent, and Plancherel gives

$$\|k_\epsilon\|_\infty \le \frac{1}{2\pi\epsilon}, \qquad \|k_\epsilon\|^2_{L^2(\mu)} = \frac{1}{4\pi\epsilon}.$$

The convolution kernel of $D_\epsilon = AB_\epsilon - B_\epsilon M^{-1}B_\epsilon MA$ is therefore

$$\begin{aligned} D_\epsilon(s,t) &= F_\hbar(s)k_\epsilon(s-t) \\ &\quad - M(t)F_\hbar(t)\int_G k_\epsilon(s-u)\overline{M(u)}k_\epsilon(u-t)\,d\mu(u). \end{aligned}$$

The inner integral is absolutely convergent by Cauchy–Schwarz, bounded in modulus by $1/(4\pi\epsilon)$. For Schwartz input and output tests the remaining double pairing is also absolutely convergent. This does not authorize expanding every Fourier integral and exchanging all oscillatory integrations at once.

Dominated convergence in the momentum spectral measure gives $B_\epsilon \to B$ strongly. Products are uniformly bounded, so $D_\epsilon \to D = AB - BM^{-1}BMA$ strongly. No unbounded chiral factor or fixed-vector holomorphic continuation is used.

For normalized f, g, put $v = MAf$ and $E_\epsilon(w) = \|(1 - H_\epsilon)w\|$. A telescoping difference and Cauchy–Schwarz yield the useful matrix-element bound

$$|\langle g, (D - D_\epsilon)f\rangle| \le E_\epsilon(f) + E_\epsilon(g) + E_\epsilon(v). \tag{5}$$

For example, moving the outer $B - B_\epsilon$ to the output test costs $E_\epsilon(g)$, since B commutes with H_ϵ and is unitary. The inner difference costs $E_\epsilon(MAf)$. All three errors tend to zero for every fixed L^2 test. One must still compute and control a matrix element of D_ϵ; its value need not be zero even if $D = 0$.

5 Gaussian reduction of the only rough heat tail

In $L^2(\mu)$ take $f = g = f_0 = \sqrt{2}\, e^{-|s|^2/(2\pi)}$, so $\|f_0\| = 1$. Then

$$E_\epsilon(f_0)^2 = \frac{2(\pi\epsilon)^2}{(1+\pi\epsilon)(1+2\pi\epsilon)}.$$

For $v(s) = r(s)f_0(s)$, $r = MF_\hbar$, define the real heat overlap $C_t = \langle v, H_t v\rangle$. In ordinary coordinates the heat kernel is $(4\pi^3 t)^{-1} e^{-|s-s'|^2/(4\pi^2 t)}$. Using $U = (s+s')/2$, $Z = s - s'$, Gaussian completion gives

$$C_t = \frac{1}{1+\pi t}\, \mathbb{E}\Big[\overline{r(U+Z/2)}\, r(U-Z/2)\Big], \tag{6}$$

where U, Z are independent centered two-dimensional Gaussians with

$$\mathrm{Cov}(U) = \frac{\pi}{2} I, \qquad \mathrm{Cov}(Z) = \frac{2\pi^2 t}{1+\pi t} I.$$

Symmetry makes the expectation real. An explicitly nonnegative form, useful near cancellation, is

$$1 - C_t = \frac{\pi t + \mathbb{E}L_t}{1+\pi t}, \qquad L_t = \frac{1}{2}|r(U+Z/2) - r(U-Z/2)|^2 \in [0,2]. \tag{7}$$

The spectral inequality $(1-a)^2 \le 1 - a^2$, $0 \le a \le 1$, proves

$$E_\epsilon(v)^2 \le 1 - C_{2\epsilon}.$$

Combining it with (5) yields the exact bound functional

$$\mathcal{E}_\epsilon = 2\sqrt{\frac{2(\pi\epsilon)^2}{(1+\pi\epsilon)(1+2\pi\epsilon)}} + \sqrt{1 - C_{2\epsilon}}. \tag{8}$$

No H^1 assumption on the vortex phase is needed. A numerical estimate of (7) is not itself a certified upper bound.

As a separate analytic control, remove F but retain $r = M$. Integrating the Gaussian bilinear phase in (6) gives $C_t = (1+2\pi t)^{-1}$. If $r = 1$, instead $C_t = (1+\pi t)^{-1}$, which recovers the exact vacuum formula above. These controls test the measure, covariance and chirp normalization.

6 Exploration, evidence and the next decision

We used four scrambled Sobol seeds, 2^{16} four-dimensional normal samples per seed, three paired cases $\hbar = .5, 1, 2$, the analytic chirp control, and five ϵ values from 10^{-2} to 10^{-6}. Prefixes $2^{12}, 2^{14}, 2^{16}$ give 240 saved records. Prefixes and settings reuse base nodes; they are not independent replications. Seeds, versions, input hashes, prefix moments and counts of large phase losses are saved.

For $\hbar = 1$, the four fine-grid estimates of (8) span:

ϵ	minimum	maximum
10^{-2}	.376212	.376429
10^{-3}	.106305	.106500
10^{-4}	.0319381	.0324646
10^{-5}	.00991042	.0101192
10^{-6}	.00311482	.00318232

At 10^{-6} the ranges for $\hbar = .5, 2$ are approximately .00307294–.00307389 and .00307308–.00307398. The chirp-only control has maximum relative deviation $2.80\,10^{-4}$ at the fine order. Thirty scalar, symbolic and control checks pass. These figures explore feasibility only. The larger replicate spread at $\hbar = 1$ is a reason to resolve rare phase-loss regions, not evidence of a certified error bar.

Next scientific calculation. Use the matched bounded kernel to compute one specified $\langle f_0, D_\epsilon f_0 \rangle$ with certified integration error, and separately bound the expectation in (7). Split off Gaussian tails and neighborhoods of the known phase singularities; their contribution is controlled by $0 \le L_t \le 2$. Only a certified modulus exceeding the sum of those errors can yield a continuous nonzero defect. If it does not, retain that outcome and refine the representation/identity comparison. The earlier finite-grid residuals and the present empirical heat estimates cannot be combined into a counterexample.

This supplies an executable route around the unbounded-factor difficulty, using known Fourier and heat-semigroup facts. It does not resolve the final pentagon, provide observer mass labels or construct the finite-G physical gluing measure.

References

[1] H. K. Kim and C. Scarinci, *A quantization of moduli spaces of 3-dimensional gravity*, 2112.13329v2, §6.3, Proposition 6.15.

[2] S. Garoufalidis and R. Kashaev, *A meromorphic extension of the 3D index*, 1706.08132v2, Appendices A–B, equations (129)–(138).

[3] S. Garoufalidis and R. Kashaev, *Quantum dilogarithms over local fields and invariants of 3-manifolds*, 2306.01331v2, Example 2.4, Remark 2.6, Definition 3.1, Lemma 3.2, Theorem 3.3.

[4] R. Kashaev, *On beta pentagon relations*, 1403.1298v1, Lemma 1 and Theorem 1.